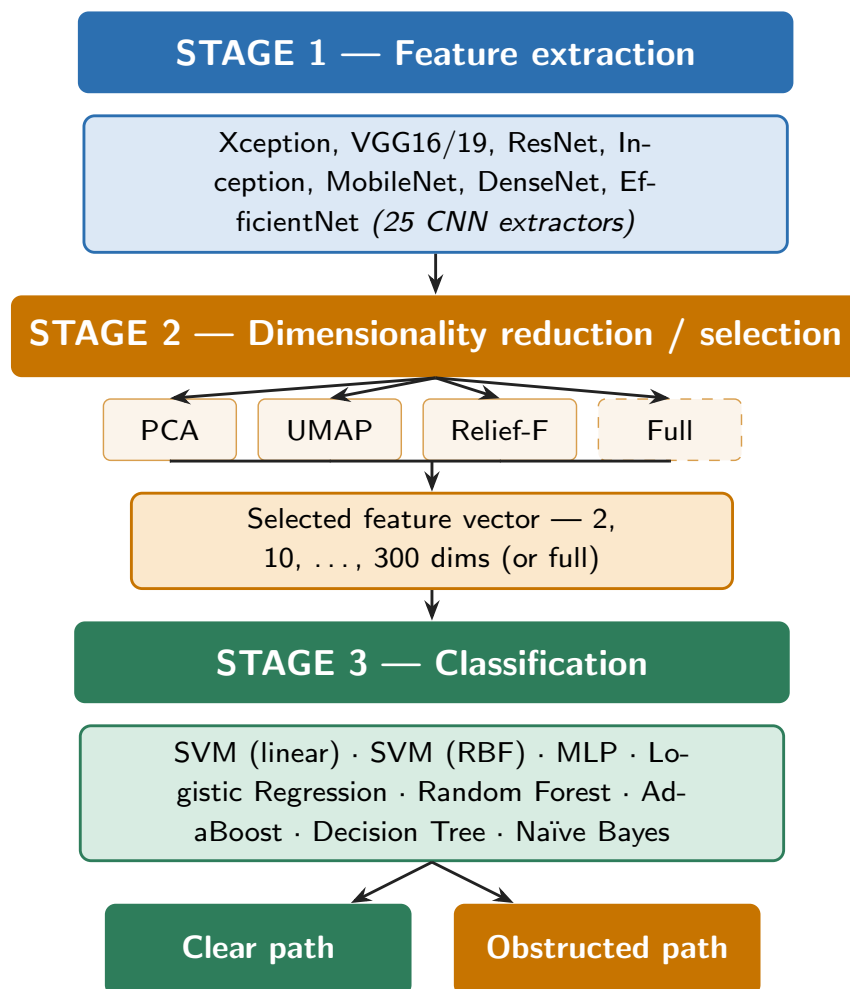


Graphical Abstract

Cross-Analysis of CNN Architectures as Feature Extractors for Obstacle Detection to Aid the Visually Impaired

Jefferson Antonio Ribeiro Passerini, Fabricio Breve



LOOCV accuracy: 0.9474 • 10-fold accuracy: 0.9412

Highlights

Cross-Analysis of CNN Architectures as Feature Extractors for Obstacle Detection to Aid the Visually Impaired

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- 25 CNN architectures evaluated as feature extractors for obstacle detection in urban navigation.
- Features extracted using transfer learning without CNN retraining.
- Dimensionality reduction boosted classifier performance and stability.
- Best accuracy of 95.59% achieved with MobileNet and ResNet50 features and a linear SVM.
- The most stable configuration across tests was EfficientNetB0 with MobileNet.
- The classification stage adds negligible computational cost (about 6.7 ms per image).

Cross-Analysis of CNN Architectures as Feature Extractors for Obstacle Detection to Aid the Visually Impaired

Jefferson Antonio Ribeiro Passerini, Fabricio Breve

Institute of Geosciences and Exact Sciences, São Paulo State University (UNESP), Rio Claro, SP, 13506-900, Brazil

Abstract

This study presents a comparative analysis of convolutional neural networks (CNNs) used as feature extractors, combined with dimensionality reduction techniques and supervised classifiers, for obstacle detection aimed at assisting visually impaired individuals. Without requiring fine-tuning or retraining, 25 CNN architectures – including EfficientNet variants up to B7 – were used as feature extractors, while a reference model based on 22 fine-tuned architectures served as a baseline. The features were reduced using PCA, UMAP, or Relief-F and classified using models such as SVM and MLP. Among the four proposed approaches, the best accuracy (95.59%) was achieved using a combination of MobileNet and ResNet50 features with a linear SVM in the 10-fold cross-validation. However, the most consistent performance across all evaluations, including the LOOCV (94.74%) and the subset of images with ground traffic signage (92.59%), was obtained with the EfficientNetB0 + MobileNet configuration. Heatmap analyses revealed challenges in classifying signage-based images, suggesting future improvements using attention mechanisms or semantic segmentation. This offline comparative study shows that CNNs used purely as feature extractors can match a fine-tuned baseline at a fraction of its training cost; the on-device deployment and evaluation of the selected model on a smartphone are left for future work.

Keywords: Assistive Technology, Deep Learning, CNN, Transfer Learning,

Email addresses: `jefferson.passerini@unesp.br` (Jefferson Antonio Ribeiro Passerini), `fabricio.breve@unesp.br` (Fabricio Breve)

1. Introduction

Visual impairments, particularly blindness and low vision, represent an escalating challenge for institutions and governments across the globe, as highlighted by the World Health Organization [1]. The rising prevalence of eye-related diseases, coupled with a decline in visual acuity, significantly impacts individuals' quality of life and necessitates substantial financial resources for their treatment and rehabilitation [1]. Currently, over 2.2 billion individuals suffer from some form of visual impairment, yet access to assistance and services for this demographic remains unequal, according to the United Nations [1, 2, 3, 4].

Individuals with visual impairments face numerous challenges in their daily lives, especially in terms of mobility and navigation. Traditionally, this population has relied on canes and guide dogs as their primary means of locomotion [5]. However, advancements in computer vision have led to the development of various navigation systems [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17]. It is important to note that many of these systems are limited by the need for expensive, bulky, and/or customized equipment [13, 14, 15, 16, 18, 19], as well as the requirement for high computational power, rendering them impractical for use on mobile devices. Additionally, some systems necessitate a network connection to a more powerful remote server [11, 20].

A comprehensive literature review by Budrionis et al. [7] revealed that computer vision tools designed for visually impaired individuals operating on smartphones frequently utilize outdated image and video processing techniques. Another literature review by Mandia et al. [6] found an increasing adoption of deep learning approaches [21, 22, 23], which has grown in tandem with advancements in computational capabilities. Nonetheless, it is impractical for users to carry high-power computing devices for vision-based assistive solutions. Consequently, there is an increasing need for a deep learning system capable of performing inferences on an edge device, such as a smartphone, without relying on network connectivity or additional accessories.

Breve and Fisher [24] proposed a framework employing Convolutional Neural Networks (CNNs), transfer learning, and semi-supervised learning

(SSL) aimed at reducing computational costs and making it feasible for implementation on smartphones without the need for extra hardware. This framework leverages a smartphone’s camera to instantly capture images of the user’s surroundings and classify them in real time, providing immediate feedback. In this study, a dataset consisting of images from various outdoor and indoor environments under different lighting conditions, floor types, and obstacles was created to facilitate classifier training. The effectiveness of several CNN architectures was assessed by adjusting the pre-trained weights from the ImageNet dataset [25].

Sanga et al. [26] developed a prototype based on the framework proposed by Breve and Fisher [24], consisting of a mobile app that captures video frames from the smartphone’s camera, sends them to a server for image classification, and then returns the results to the user. Furthermore, Breve [27] significantly expanded the initial study by incorporating new CNN models, enhancing the cross-validation process, and applying pre-processing functions to improve image preparation. These improvements resulted in enhanced performance in most analyzed cases.

Building on the work of Breve and Fisher [24] and Breve [27], this paper does not seek to train a better classifier, but to systematically assess whether CNNs used *purely as feature extractors* – without any fine-tuning – can match a fine-tuned baseline when combined with dimensionality reduction/selection and classical supervised classifiers. To the best of our knowledge, a cross-analysis of this breadth – spanning individual CNNs, combinations of CNNs, three reduction/selection methods, and eight classifiers – has not been reported for the obstacle-detection task addressed here. The key contributions are as follows:

1. A systematic cross-analysis of 25 CNN feature extractors – and their pairwise combinations – three dimensionality reduction/selection algorithms (PCA, UMAP, Relief-F), and eight supervised classifiers, all evaluated without training any CNN.
2. Evidence that the best configurations are statistically equivalent (Friedman test with Nemenyi post-hoc) to a fine-tuned, 22-architecture reference model, at a fraction of its computational cost.
3. A comparison of four feature-combination strategies (full features, single-CNN reduction, and two concatenation schemes), identifying which of them generalise reliably under leave-one-out validation.
4. An interpretability-based error analysis (Grad-CAM) that reveals a systematic failure mode – the confusion between obstacles and painted ground signage – and points to concrete directions for future improvement.

The remainder of the paper is organized as follows: Section 2 presents related work on assistance for visually impaired individuals. Section 3 introduces the dataset used in this study. Section 4 outlines the CNN architectures and algorithms utilized. Section 5 describes the methodology employed. Section 6 discusses the experimental results and analyses, comparing these models’ performance on the dataset and exploring deficiencies in the proposed approach. Finally, conclusions are summarized in Section 7.

2. Related Works

Several initiatives have been undertaken to incorporate computer vision into assistance for individuals with visual disabilities. Mandia et al. [6] conducted a comprehensive analysis of current vision-based assistive solutions for visually impaired people.

This analysis primarily focuses on systems utilizing cameras and summarizes the sensors, image processing algorithms, and communication protocols employed. Additionally, the study addresses the use of acoustic output devices, RFID, and GPS, along with cameras. The transition from traditional image processing techniques to the application of deep learning in assisting visually impaired individuals is highlighted. The paper concludes that the literature has not yet fully optimized deep learning models for edge devices.

Budrionis et al. [7] provide an overview of recent prototypes of electronic travel aids (ETAs) that utilize smartphones to assist visually impaired individuals with orientation, navigation, and mobility. The authors conduct a systematic review of scientific achievements in this field and compare various smartphone-based ETA prototypes. The meta-analysis identified some attempts to employ cutting-edge computer vision methods based on deep neural networks. The study contrasts these findings with a survey conducted with blind experts to reveal a significant discrepancy between user needs and academic development in the field. The authors conclude that the development of affordable smartphone-based ETAs is crucial for visually impaired people in low-income countries and highlight the need for further research to address the identified gaps.

Islam et al. [5] analyze the development of walking assistants for visually impaired individuals and highlight recent advancements, as well as their advantages and limitations. The authors aim to provide a comprehensive view of the current state of walking assistants and identify areas for future development in sensors, computer vision, and smartphone-based technology.

In turn, Kuriakose et al. [8] propose a scene recognition model based on EfficientNet-Lite for use in a smartphone application that assists in the navigation of blind and visually impaired people. The model is trained and tested using a custom dataset of indoor and outdoor scenes. Additionally, a proof-of-concept prototype of the application was developed on the Android platform.

Bai et al. [10] introduced a wearable assistive device for visually impaired (VI) individuals that aids in navigation and object recognition in both indoor and outdoor environments. The device consists of an RGB-D camera, an inertial measurement unit, a smartphone, and an audio module. It employs a CNN-based object recognition system for perception and navigation. The system provides semantic information about the environment and interacts with the user through audio.

Jiang et al. [11] proposed a wearable system that utilizes stereo vision. The system leverages binocular vision sensors to capture images and selects the best images based on stereo image quality assessment. The selected images are then sent to the cloud for processing via a convolutional neural network (CNN), and the resulting information is returned to the user to assist in decision-making.

Paul et al. [12] propose a system composed of a camera, GPS, infrared and light sensors connected to a microprocessor (Raspberry Pi) to process and transmit information about the user's environment. The camera captures images, while the microprocessor uses image processing techniques to analyze the images and identify objects and obstacles. The control unit then transmits this information to the user via audio output. The authors did not specify which processing techniques were used, mentioning only the use of the OpenCV library.

Hoang et al. [13] created a system that uses a Kinect sensor camera mounted on a belt to capture and analyze the surrounding environment. This system identifies obstacles and communicates this information to the user through audio feedback. To process the captured information, it is necessary to carry a laptop in a backpack.

In 2013, Tapu et al. [28] introduced a real-time obstacle detection and classification system that made use of videos captured by a smartphone's camera. They developed a set of techniques that included tracking, motion estimation, and clustering. Four years later, in 2017, Tapu et al. [14] presented a more advanced framework based on convolutional neural networks (CNNs) to detect, track, and recognize objects in outdoor environments.

However, to run this system, it is necessary to use a laptop computer, which must be carried in a backpack as a processing unit.

Lin et al. [20] introduced a guidance system that makes use of a smartphone. The system incorporates convolutional neural networks (CNNs) for object recognition but requires a desktop server with a GPU and the unified computing device architecture (CUDA) to handle the computationally intensive task of object recognition. Although the system has an offline option, it is limited to recognizing faces and stairs.

Kumar and Meher [29] introduced an object recognition system that combines the use of CNN (Convolutional Neural Network) and RNN (Recurrent Neural Network). This system is capable of identifying common objects found in indoor environments and their colors, providing auditory feedback to the user. In contrast, Saffoury et al. [15] proposed a system that uses a smartphone and a laser pointer. The system employs laser triangulation to establish a collision prevention protocol and provides auditory feedback to the user.

Poggi and Mattoccia [16] conceived a wearable device composed of a pair of glasses equipped with a custom RGBD sensor and onboard processing performed by an FPGA, a glove with micro-motors for tactile feedback, a portable battery, a bone conduction headset, and a smartphone. This system utilizes deep learning techniques to semantically classify detected obstacles. In a previous work, Poggi et al. [18] introduced a similar system aimed at recognizing pedestrian crosswalks.

Rizzo et al. [19] present a fusion framework that combines signals from a stereo camera and an infrared sensor to detect obstacles using a CNN at multiple scales. They plan to incorporate this framework into a wearable vest.

On the other hand, Islam and Sadi [30] used a CNN to detect potholes in the path and achieved impressive results. However, it is important to note that they used separate datasets for “path potholes” and “non-path potholes”, with the latter including images of roads captured at a wider angle. This raises the possibility that the network learned style differences that are not relevant to the task at hand, simplifying its work.

3. Dataset

The VIA dataset [24] comprises 342 images distributed across two categories: 175 “clear path” images and 167 “obstructed path” images. The



Figure 1: Examples from the VIA dataset: (a) “clear path” category; and (b) “path with obstacles” category.

photos were taken using a smartphone camera and resized to 750×1000 pixels. The smartphone was held at chest height and tilted at an angle of 30° to 60° to capture several meters of the path ahead, including areas not reached by a standard cane. Despite its modest size, this dataset encompasses a variety of indoor and outdoor environments with different types of flooring, including dry and wet surfaces, natural and artificial lighting conditions, as well as obstacles such as stairs, trees, holes, animals, and traffic cones. Examples of these images can be seen in Fig. 1.

4. CNN Architectures and Algorithmic Approaches

This section presents the CNN architectures explored in this study, as well as the feature reduction/selection algorithms and classifiers employed herein.

Table 1 displays the 22 evaluated architectures, along with some of their features and literature references. The original CNN architectures were utilized with their existing structures and weights for ImageNet classification

[25] through transfer learning, except for the dense classification layers, which were removed.

Table 1: CNN Architectures, Input Sizes, and Key Characteristics.

CNN	Img Res.	Last Conv. (out)	Train. Params	Ref.
DenseNet121	224x224	7x7x1024	8.1M	[31]
DenseNet169	224x224	7x7x1664	14.3M	[31]
DenseNet201	224x224	7x7x1920	20.2M	[31]
EfficientNetB0	224x224	7x7x1280	5.3M	[32]
EfficientNetB1	240x240	8x8x1280	7.9M	[32]
EfficientNetB2	260x260	9x9x1280	9.2M	[32]
EfficientNetB3	300x300	10x10x1536	12.3M	[32]
EfficientNetB4	380x380	12x12x1792	19.5M	[32]
Inception-V3	229x229	8x8x2048	23.9M	[33]
Inception-ResNet-V2	229x229	8x8x1536	55.9M	[34]
MobileNet	224x224	7x7x1024	4.3M	[35]
MobileNetV2	224x224	7x7x1280	3.5M	[36]
NASNetMobile	224x224	7x7x1056	5.3M	[37]
ResNet50	224x224	7x7x2048	25.6M	[38]
ResNet50V2	224x224	7x7x2048	25.6M	[39]
ResNet101	224x224	7x7x2048	44.7M	[38]
ResNet101V2	224x224	7x7x2048	44.7M	[39]
ResNet152	224x224	7x7x2048	60.4M	[38]
ResNet152V2	224x224	7x7x2048	60.4M	[39]
VGG16	224x224	7x7x512	138.4M	[40]
VGG19	224x224	7x7x512	143.7M	[40]
Xception	299x299	10x10x2048	22.9M	[41]

In addition to the 22 architectures listed in Table 1, which compose the reference model and are also employed as feature extractors, the EfficientNetB5, EfficientNetB6, and EfficientNetB7 variants [32] were used exclusively as feature extractors in the proposed approaches. Because using a CNN as a feature extractor relies solely on transfer learning and does not require fine-tuning, these larger variants could be evaluated in this role despite the hardware constraints that prevented their training as part of the reference model.

For dimensionality reduction, the Principal Component Analysis (PCA) [42] and Uniform Manifold Approximation and Projection (UMAP) [43] algorithms were used. Feature selection was also conducted using the Relief-F selection model [44].

For the model classification tests, the following algorithms were employed: Decision Tree [45], Support Vector Machine - SVM with RBF kernel [46], SVM with linear kernel [47], Multilayer Perceptron - MLP [48], Logistic Regression [49], Random Forest [50], Adaboost [51], and Gaussian Naïve Bayes [52].

5. Proposed Methodology

For the comparative evaluation of results, the methodology adopted by Breve and Fisher [24] was defined as the reference model, and different approaches utilizing CNNs, feature reduction/selection algorithms, and classifiers were proposed in the study for classifying the dataset into clear or obstacle-laden paths for visually impaired users.

For the training and testing of reference models, a computer equipped with an AMD Ryzen 9 3900X processor, 64 GB of RAM, Windows 11 operating system, and an NVIDIA GeForce RTX 2060 SUPER graphics card with 8GB of memory and 2,176 CUDA cores was used.

The tests of the proposed approaches were conducted on a notebook with an Intel Core i7-9750H 2.6GHz processor, 32GB of RAM, running a 64-bit POP OS Linux operating system with kernel version 5.17.5. The models were developed in Python language, version 3.9.12, executed on the Anaconda platform where Spyder IDE version 4.1.4 was used.

The tests of the proposed approaches and reference CNNs were carried out using k -fold cross-validation ($k = 10$). To verify the test results, the accuracy (acc) was measured at each execution of the classifier. For statistical validation of the results and ranking of the models within each approach, the non-parametric Friedman test was used, through the statistical software IBM SPSS Statistics, version 20.0.0. As the models were evaluated on the same folds (repeated measures), the Friedman test is appropriate both to verify the existence of statistically significant differences and to rank the models. The complete per-fold results, spreadsheets, and statistical outputs of all experiments are publicly available at <https://github.com/jeffersonarpasserini/Phd-Partial-Results>.

5.1. Baseline CNN

As a performance benchmark for classifying images into clear paths or paths with obstacles, the training of 22 convolutional network architectures

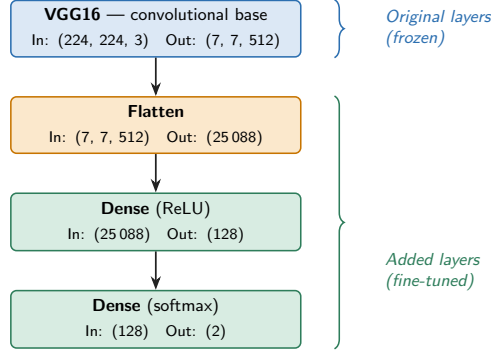


Figure 2: Layers added to the original CNN Architecture (Example: VGG16).

was conducted. The following convolutional neural networks were utilized as presented in Table 1.

Following the approach proposed by Breve and Fisher [24], from the architectures of each of the convolutional networks, the original pre-trained layers were retained, and new layers were added to the model. These new layers were trained with images from the dataset for the study (fine-tuned), as can be seen in Fig. 2.

The reference model was evaluated based on the proposed tests to validate the CNN with the best performance for the problem of obstacle identification in the dataset.

5.2. Approaches Proposed

The approaches proposed in this study were structured into three stages:

1. Feature extraction through CNNs.
2. Feature reduction or selection.
3. Classification into clear path or path with obstacles.

For the feature extraction process, the CNNs were not specifically trained for the problem proposed by the work but utilized Transfer Learning with pre-trained parameters from the ImageNet dataset [25]. Thus, information from the last flattened convolutional layer (flatten layer) was used as the feature vector without employing a pooling layer, as illustrated in Fig. 3 with an example from the VGG16 CNN.

The second stage of the proposed method involves the use of a model for reduction and/or selection of features or the passing through of all extracted

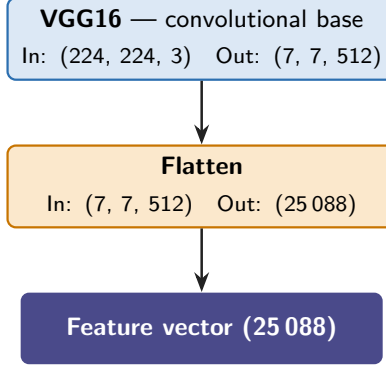


Figure 3: Example of the feature extraction process (VGG16).

features. The feature reduction algorithms applied were Principal Component Analysis (PCA) [42], Uniform Manifold Approximation and Projection (UMAP) [43], and the feature selection algorithm used was the Relief-F model [44]. In this phase, when the algorithm was applied, feature vectors of different sizes were generated: 2, 10, 20, 30, 40, 50, 75, 100, 150, 200, 250, and 300 features, reducing the original dimensionality of the output from the convolutional neural network (See: Fig. 4).

In the third stage, classification is performed using the feature sets produced in stage 2, with the classifiers introduced previously.

Fig. 4 summarizes the resources used for analyzing the proposed model and demonstrates combined approaches of CNN architectures in feature extraction in stage 1 of this model. It can also be concluded that there are many possible combinations among stages 1, 2, and 3, thereby structuring four study approaches that we will denote as A, B, C, and D. Tests were initially conducted within each set of the proposed approaches. Within each group of approaches, the model that achieved the highest statistical relevance for the problem was selected.

With the selected models, an analysis was conducted to compare the proposed approaches with the reference model and to produce a ranking among the techniques. Subsequently, a study was also carried out to determine the behavior of the selected models in relation to the classification process for an individual image. For this purpose, leave-one-out cross-validation tests were conducted, as well as an evaluation of the convolutional neural networks through a heatmap analysis.

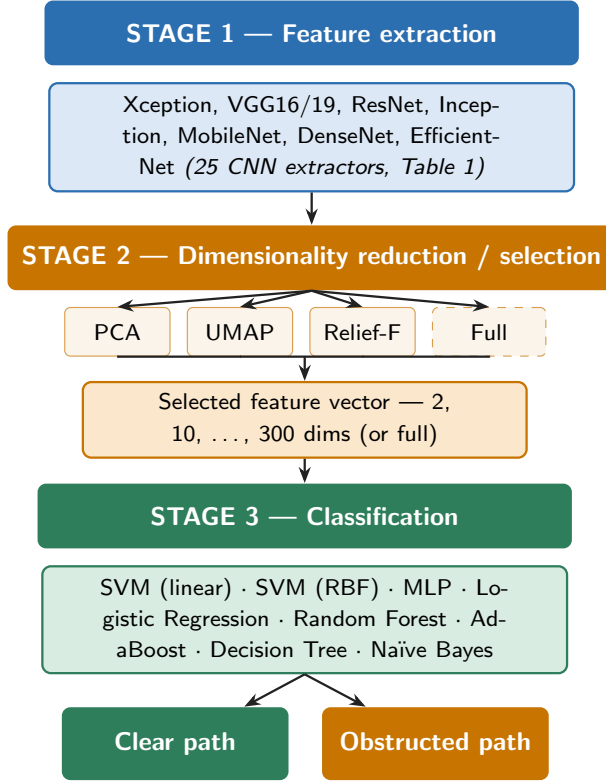


Figure 4: Combination of possible algorithms at each stage.

In Approach A, feature extraction was utilized for all the individual CNNs - Fig. 5a and their combinations - Fig. 5b, using the CNN combinations described in this section, leveraging the full output of features extracted by the convolutional network without the process of dimensionality reduction, and feeding the data into each of the classifiers used. This approach generated 272 different technique combinations.

In this approach, when combinations of CNNs were used for feature extraction, the output vectors of each of the convolutional networks are concatenated and applied to the classifier (See: Fig. 5b).

In Approach B (Fig. 6), only the individual CNNs were utilized, with the output of features extracted by the convolutional network undergoing a dimensionality reduction process through PCA, UMAP, or Relief-F, where each of the models generated selected feature vectors of sizes 2, 10, 20, 30, 40,

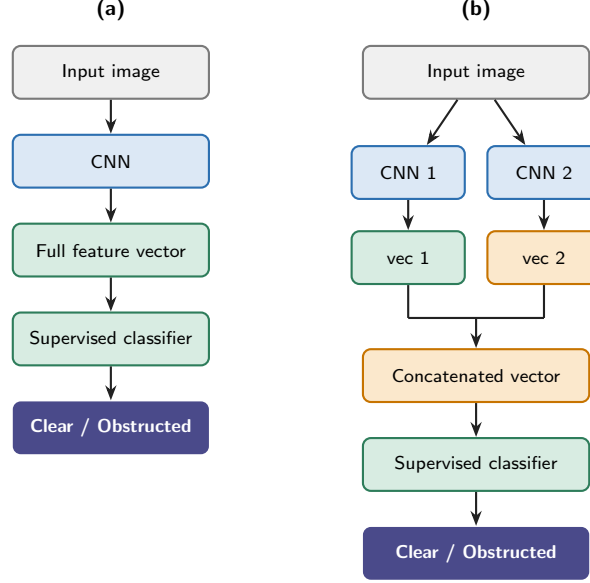


Figure 5: Proposed methodology in Approach A: (a) utilization of the output from only 1 CNN, (b) utilization of the output from 2 CNNs and the concatenation process of the output.

50, 75, 100, 150, 200, 250, and 300. The output vectors generated from the dimensionality reduction process were applied to the 8 classifiers used in this study. This process resulted in 7,200 technique combinations for Approach B.

In Approach C (Fig. 7), combinations of feature vectors extracted from the CNNs were used, as described in this section.

This process involves concatenating the output vectors from each of the applied CNNs into a single vector before subjecting the data to the dimensionality reduction process. The dimensionality reduction utilized the previously described techniques (PCA, UMAP, and Relief-F), where output vectors of sizes 2, 10, 20, 30, 40, 50, 75, 100, 150, 200, and 300 were generated. Each of these vectors was applied to the 8 classifiers, resulting in a total of 2,592 possible technique combinations for Approach C.

For Approach D (Fig. 8), combinations of feature vectors extracted from the CNNs were also utilized, altering the vector creation process for the classifier. In this approach, each of the utilized CNNs performs feature extraction, and its output vector is subjected to the dimensionality reduction process

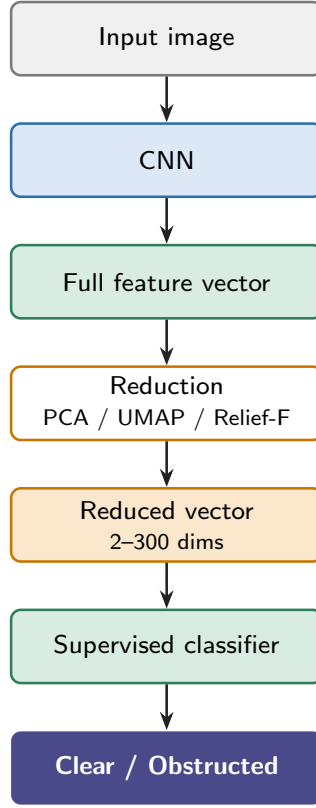


Figure 6: Proposed methodology in Approach B.

using one of the proposed techniques, generating the selected feature vector of the desired size (2, 10, 20, 30, 40, 50, 75, 100, 150, 200, or 300). The process occurs individually for each of the CNNs, and only after obtaining the feature vector from each CNN, the feature vector for classification is generated from the concatenation of both. The final vector for classification is subjected to each of the 8 classifiers, resulting in 2,592 possible combinations of the employed techniques.

6. Results

The tests were conducted using k-fold cross-validation ($k=10$), and the non-parametric Friedman test was performed to determine the best approach for the proposed problem; this test uses the median as a measure of perfor-

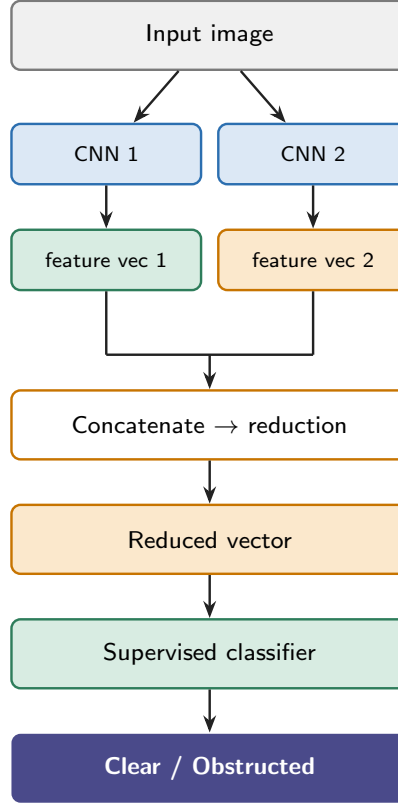


Figure 7: Proposed methodology in Approach C.

mance.

Given the results presented in the CNN tests, the Null Hypothesis (H0) is that there are no statistically significant differences between their performances, and (H1) such differences exist, making it possible to determine a model suitable for the studied problem.

Upon applying the test to the reference model, the Friedman test indicated a statistically significant difference among the architectures examined [$X^2(21) = 75.062; p < 0.0001$], with the significance level set at $p < 0.05$, leading to the rejection of the null hypothesis and enabling the ranking of the architectures. The descriptive outcomes of these performance assessments, which are based on the accuracy achieved and the classification ranking of each studied CNN architecture, are detailed in Table 2.

For the proposed Approach A, the tests followed the same standards,

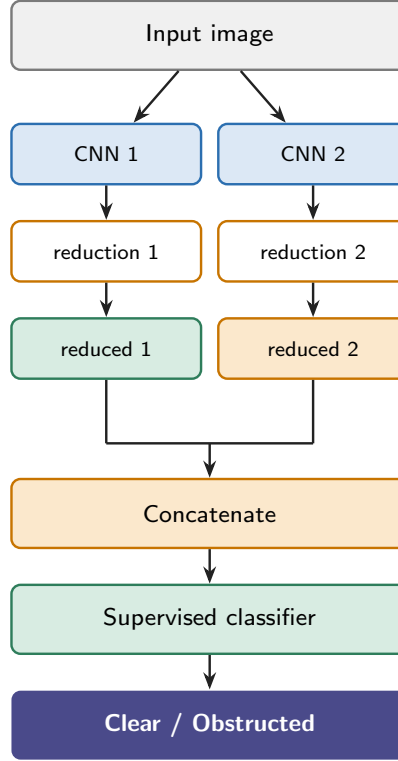


Figure 8: Proposed methodology in Approach D.

wherein the Friedman test made it possible to verify that there is a statistically significant difference between the techniques studied [$X^2(271) = 1,731.191; p < 0.0001$]. The significance level of $p < 0.05$ determines the rejection of the null hypothesis. In Table 3, one can observe the descriptive results of the performance tests and the corresponding Friedman ranking. Due to the large number of combinations of techniques, the twenty highest-ranked techniques were presented.

The proposed model described in Approach B utilized individual CNNs combined with dimensionality reduction models that resulted in output vectors of sizes 2, 10, 20, 30, 40, 50, 75, 100, 150, 200, 250, and 300. These were then applied to 8 studied classifiers, resulting in a considerable number of technique combinations (7200). Thus, the Friedman test was applied to each of the CNNs individually, resulting in 288 technique combinations for each, thereby selecting the best-performing proposal for each CNN.

Table 2: CNN Architectures and Their Performance Ranking

CNN	Rank	Q1 25%	Q2 50%	Q3 75%	IQR
DenseNet201	18.25	0.9118	0.9412	0.9416	0.0298
EfficientNetB0	16.65	0.8849	0.9118	0.9412	0.0563
InceptionV3	16.65	0.9045	0.9131	0.9486	0.0441
EfficientNetB3	14.55	0.8824	0.9118	0.9210	0.0386
Xception	14.20	0.8750	0.8988	0.9265	0.0515
EfficientNetB2	13.90	0.8529	0.8971	0.9412	0.0883
MobileNet	13.45	0.8677	0.8988	0.9412	0.0735
EfficientNetB1	12.80	0.8529	0.8971	0.9143	0.0614
DenseNet121	12.75	0.8824	0.8841	0.9118	0.0294
VGG16	12.60	0.8456	0.8824	0.9412	0.0957
DenseNet169	12.20	0.8561	0.8824	0.9124	0.0564
ResNet50V2	12.15	0.8529	0.8824	0.9210	0.0681
ResNet50	11.70	0.8456	0.8824	0.9416	0.0961
VGG19	10.80	0.8456	0.8677	0.9192	0.0736
InceptionResnetV2	10.30	0.8235	0.8677	0.9210	0.0975
EfficientNetB4	10.15	0.8456	0.8529	0.9196	0.0740
ResNet152	9.55	0.8200	0.8677	0.9118	0.0918
ResNet152V2	8.20	0.7941	0.8530	0.9124	0.1183
ResNet101V2	7.35	0.8235	0.8529	0.8824	0.0589
ResNet101	7.10	0.8235	0.8382	0.8603	0.0368
MobileNetV2	5.00	0.7550	0.8235	0.8603	0.1053
NasNetMobile	2.80	0.5107	0.6176	0.6912	0.1805

From this initial selection within Approach B, the most relevant technique combinations for this approach were analyzed. The Friedman test yielded [$X^2(24) = 49.546; p < 0.0001$], confirming statistically significant differences between the employed techniques and enabling the ranking shown in Table 4, which identified the best technique within Approach B in this study.

As can be seen in Table 4, the technique involving the CNN MobileNet with dimensionality reduction via PCA, selecting a vector of 40 features, and the SVM classifier with an RBF kernel, achieved the best performance for this approach.

In Approach C, combinations of feature vectors extracted from the CNNs were utilized. This process involves concatenating the output vectors from

Table 3: Proposed Model – Approach A: Friedman Test ranking and descriptive analysis performance

Approach A	Rank	Q1 25%	Q2 50%	Q3 75%	IQR
EfficientNetB0+ MobileNet +LinearSVM	234.75	0.9118	0.9412	0.9706	0.0588
MobileNet+ResNet50_ LinearSVM	230.75	0.9273	0.9412	0.9708	0.0435
EfficientNetB2_ LinearSVM	228.60	0.9137	0.9412	0.9706	0.0569
EfficientNetB0+ MobileNet_Logistic	228.25	0.9053	0.9412	0.9706	0.0653
ResNet101+MobileNetV2 _LinearSVM	226.65	0.9118	0.9143	0.9706	0.0588
ResNet50_Logistic	226.60	0.9053	0.9421	0.9706	0.0653
ResNet50_LinearSVM	225.20	0.9053	0.9274	0.9706	0.0653
Xception+ResNet50_ Logistic	225.20	0.9053	0.9274	0.9706	0.0653
EfficientNetB3_ LinearSVM	224.65	0.9118	0.9412	0.9416	0.0298
MobileNetV2_LinearSVM	222.60	0.9118	0.9421	0.9706	0.0588

each of the applied CNNs into a single vector before subjecting the data to the dimensionality reduction process. For dimensionality reduction, previously utilized techniques such as PCA, UMAP, and Relief-F were employed, generating selected feature vectors of sizes 2, 10, 20, 30, 40, 50, 75, 100, 150, 200, and 300; each vector was applied to the 8 studied classifiers.

This approach resulted in 2592 possible technique combinations. Accordingly, the Friedman test was conducted to determine the best technique per set of CNNs used in the feature extraction process, and finally, tests were applied to define the best technique for this approach. Separation by CNN sets created analysis groups of 288 different techniques.

From this initial selection within Approach C, the most relevant technique combinations for this approach were analyzed. The Friedman test resulted in $[X^2(8) = 22.989; p < 0.003]$, confirming statistically significant differences between the employed techniques. The corresponding ranking

Table 4: Proposed Model – Approach B: Friedman Test ranking and descriptive analysis performance

Approach B	Rank	Q1 25%	Q2 50%	Q3 75%	IQR
MobileNet_PCA_40_ SVM(RBF)	18.25	0.9118	0.9421	0.9706	0.0588
EfficientNetB1_PCA_300_ SVM (linear)	17.30	0.9339	0.9412	0.9429	0.0091
MobileNetV2_PCA_40_ SVM(RBF)	17.10	0.9118	0.9412	0.9498	0.0380
EfficientNetB2_PCA_250_ Logistic	16.20	0.8824	0.9412	0.9706	0.0882
DenseNet121_PCA_300_ Logistic	15.90	0.8824	0.9278	0.9706	0.0882
EfficientNetB0_PCA_300_ Logistic	15.70	0.9118	0.9265	0.9416	0.0298
ResNet101V2_PCA_30_ SVM (linear)	15.50	0.8849	0.9118	0.9706	0.0857
DenseNet201_PCA_300_ Logistic	15.35	0.8824	0.9278	0.9486	0.0662
EfficientNetB3_PCA_250_ Logistic	15.30	0.9118	0.9118	0.9429	0.0311
ResNet101_PCA_300_ Logistic	14.60	0.9053	0.9118	0.9412	0.0359

(Table 5) identified the best technique within Approach C.

As can be observed in Table 5, the technique involving the CNN MobileNet+ ResNet50 with dimensionality reduction via PCA, selecting a vector of 300 features, and the SVM classifier with a linear kernel, achieved the best performance for this approach.

In Approach D, the same conditions as Approach C apply, with the modification being in the construction of the feature vector where each of the CNNs undergoes a dimensionality reduction process before the merging of the output vectors. This approach resulted in 2592 possible technique combinations. Consequently, the Friedman test was conducted to determine the best technique per set of CNNs used in the feature extraction process, and ultimately, tests were applied to define the best technique for this approach.

Table 5: Proposed Model – Approach C: Friedman Test ranking and descriptive analysis performance

Approach C	Rank	Q1 25%	Q2 50%	Q3 75%	IQR
MobileNet+ResNet50_ PCA_300_SVM(linear)	7.20	0.9339	0.9559	0.9786	0.0447
EfficientB0+MobileNet_ PCA_300_SVM(linear)	6.45	0.9412	0.9412	0.9498	0.0086
ResNet101+DenseNet121_ PCA_300_Logistic	5.30	0.8849	0.9412	0.9706	0.0857
MobileNet+ResNet101_ PCA_300_Logistic	5.00	0.9118	0.9412	0.9416	0.0298
EfficientNetB1+ EfficientNetB5 _PCA_300_Logistic	4.85	0.9045	0.9278	0.9498	0.0454
ResNet101+DenseNet169_ PCA_40_SVM (RBF)	4.80	0.9045	0.9143	0.9486	0.0441
Xception+ResNet50_ PCA_300_Logistic	4.75	0.8824	0.9265	0.9498	0.0674
ResNet101+MobileNetV2_ PCA_300_Logistic	4.30	0.9053	0.9265	0.9416	0.0363
VGG16+VGG19_ ReliefF_300_SVM (RBF)	2.35	0.8397	0.8841	0.9412	0.1015

Separation by CNN groups created analysis groups of 288 different techniques.

From this initial selection within Approach D, the most relevant technique combinations for this approach were analyzed. The Friedman test resulted in $[X^2(8) = 13.266; p < 0.103]$, indicating that there are no statistically significant differences between the employed techniques. The corresponding ranking (Table 6) identified the best-positioned technique within Approach D.

As can be observed in Table 6, the technique involving the CNN MobileNet+ ResNet50 with dimensionality reduction via PCA, selecting a vector of 100 features, and the SVM classifier with an RBF kernel, achieved the best performance for this approach.

In Table 7, the descriptive performance analysis of the selected approaches

Table 6: Proposed Model – Approach D: Friedman Test ranking and descriptive analysis performance

Approach D	Rank	Q1 25%	Q2 50%	Q3 75%	IQR
MobileNet+ResNet50 _PCA_100_ SVM(RBF)	5.95	0.9045	0.9278	0.9708	0.0663
EfficientNetB0+MobileNet _PCA_200_SVM (RBF)	5.90	0.9045	0.9412	0.9706	0.0662
ResNet101+MobileNetV2 _PCA_75_SVM (RBF)	5.65	0.8761	0.9412	0.9708	0.0947
ResNet101+DenseNet121 _PCA_75_SVM (RBF)	5.55	0.8981	0.9412	0.9706	0.0725
ResNet101+DenseNet169 _PCA_75_SVM (RBF)	5.20	0.8971	0.9412	0.9412	0.0441
EfficientNetB1+ EfficientNetB5 _PCA_300_Logistic	5.15	0.9045	0.9278	0.9412	0.0368
MobileNet+ResNet101 _PCA_75_SVM (RBF)	4.50	0.8971	0.9412	0.9416	0.0445
Xception+ResNet50 _PCA_40_SVM (RBF)	4.25	0.8849	0.9118	0.9412	0.0563
VGG16+VGG19_ReliefF _300_SVM (RBF)	2.85	0.8397	0.8971	0.9412	0.1015

from previous tests and the reference model can be seen. The null hypothesis is that the reference model, the CNN DenseNet201 [31], is the best technique for classifying images as either clear path or with obstacles (H0). If any studied approach performs equally to or better than the reference model (without statistically significant differences), then hypothesis (H1) prevails.

Once the most relevant technique combinations for each of the proposed approaches and the reference model were defined, the Friedman test was applied to compare and rank them, yielding $[X^2(4) = 3.477; p = 0.481]$. As $p > 0.05$, there are no statistically significant differences between the studied approaches and the reference model. This result is consistent with hypothesis H1: the proposed approaches classify the images on par with the reference model, without the need for training a convolutional neural

Table 7: Descriptive Performance Analysis of the Selected Techniques in the studied approaches and the reference model.

Proposed Model	Appr.	Q1 25%	Q2 50%	Q3 75%	IQR
DenseNet201	Ref.	0.9118	0.9412	0.9416	0.0298
EfficientNetB0+MobileNet	A	0.9118	0.9412	0.9706	0.0588
MobileNet PCA_40_ SVM (RBF)	B	0.9118	0.9421	0.9706	0.0588
MobileNet+ResNet50 PCA_300_SVM (linear)	C	0.9339	0.9559	0.9786	0.0447
MobileNet+ResNet50 PCA_100_SVM (RBF)	D	0.9045	0.9278	0.9708	0.0663

network. In Table 8, the ranked techniques can be observed from the highest to the lowest score, as a result of the Friedman test.

Table 8: Friedman Test for the Selected Approaches and the Reference Model

Proposed Model	Friedman (Rank)
Approach C:	
MobileNet+ResNet50 PCA_300_SVM (linear)	3.70
Approach A: EfficientNetB0+MobileNet SVM (linear)	2.95
Reference Model: DenseNet201	2.90
Approach B: MobileNet PCA_40_SVM (RBF)	2.80
Approach D: MobileNet+ResNet50 PCA_100_SVM (RBF)	2.65

In the ranking, it is observed that Approach C, utilizing the CNN MobileNet and CNN ResNet50 with the PCA technique for reducing the dimensionality of the extracted data to 300 features, and the SVM classifier with a linear kernel, was ranked highest.

Since the omnibus Friedman test was already non-significant ($p = 0.481$), the Nemenyi post-hoc procedure is reported here not to detect differences but to quantify and visualise the magnitude of the pairwise gaps relative to the critical difference. For $k = 5$ models compared over $N = 10$ folds, the critical difference at $\alpha = 0.05$ is $CD = q_\alpha \sqrt{k(k+1)/(6N)} = 2.728 \sqrt{30/60} = 1.93$. Even the largest difference between the average Friedman ranks –

Approach C (3.70) and Approach D (2.65), equal to 1.05 – falls well below the critical difference. Therefore, no statistically significant difference was detected between the proposed approaches and the reference model. Fig. 9 presents the corresponding critical-difference diagram.

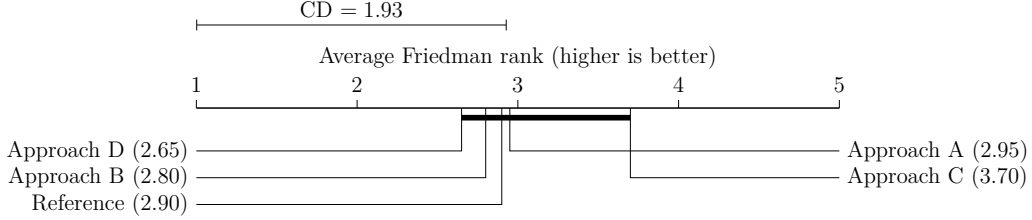


Figure 9: Critical-difference diagram (Nemenyi post-hoc, $\alpha = 0.05$, $CD = 1.93$) for the four proposed approaches and the reference model. The horizontal axis spans the full rank range (1–5), and the bracket at the top shows the critical difference to scale: because the entire spread of the five models (2.65–3.70, a range of only 1.05) is far smaller than the CD, all of them are linked by a single bar, denoting no statistically significant difference among them.

Despite the lack of statistical significance, the reference model imposes a much higher computational burden, as it requires CNN training, whereas the proposed approaches do not involve training a convolutional neural network, only the classifier.

Given the small size of the dataset, a leave-one-out cross-validation (LOOCV) test was chosen for each of the selected approaches, where each sample is used as the test set and the remaining ($N - 1$) samples are used as the training set.

In the test for each image, features were extracted from the training group and the test image, along with the processes of dimensionality reduction and classifier training. In Table 9, the performance of the approaches in the conducted test can be observed.

In Table 9, it is observed that Approach A was the only one to achieve similar performance in both tests, whereas approaches B, C, and D showed a marked reduction in accuracy when the images were tested individually (falling from 0.94–0.96 under 10-fold cross-validation to 0.84–0.87 under LOOCV). This contrast is itself a relevant finding: only the full-feature approach (Approach A) and the fine-tuned reference model remained stable across both validation protocols, indicating that the dimensionality-reduction approaches – although competitive under 10-fold cross-validation – do not

Table 9: Accuracy in the Leave-One-Out Cross-Validation Test

Proposed Model	LOOCV Accuracy	10-fold Accuracy
Approach A: EfficientNetB0+MobileNet SVM (linear)	0.9474	0.9412
Approach B: MobileNet PCA_40_SVM (RBF)	0.8713	0.9421
Approach C: MobileNet+ResNet50 PCA_300_SVM (linear)	0.8421	0.9559
Approach D: MobileNet+ResNet50 PCA_100_SVM (RBF)	0.8509	0.9278

generalise as reliably at the level of an individual image, a key consideration for a system expected to classify each frame independently.

From this test, images that most frequently exhibited prediction errors with the tested approaches were tabulated. Out of the 342 images used in the tests, 100 images had at least one prediction error in some of the approaches, equating to 29.23%. Of these 100 images, in 6 images all 4 approaches made a wrong prediction, in 12 images at least 3 approaches incurred an error, in 25 images at least 2 approaches made a wrong prediction, and in 57 images at least one approach erred in classification. Of the 100 images with erroneous classification, 51 represented paths with obstacles and 49 paths without obstacles.

Of the 6 images with errors in all approaches, 2 are of paths without obstacles as can be seen in Fig. 10. The demonstrated images are overlaid with a heatmap produced by a technique proposed by Selvaraju et al. [53], where a visual explanation is provided through the mapping of CNN activation.

Among the 6 images with errors in all approaches, 4 were of paths with obstacles, as can be seen in Fig. 11.

In the images shown in Fig. 10 (path without obstacles), it appears that the CNNs were activated by changes in ground texture or by the absence of paint on a pedestrian crossing, which ended up being characterized as an obstacle, leading to a classification error. In Fig. 11, the same phenomenon occurs: the heat areas seem to emphasize changes in texture.

Among the images with 3 errors, some examples of images depicting paths without obstacles can be seen in Fig. 12.

Apparently, in the image clear.004, there was a focus by the CNNs on the leaves scattered on the ground, and in the image clear.105, especially in

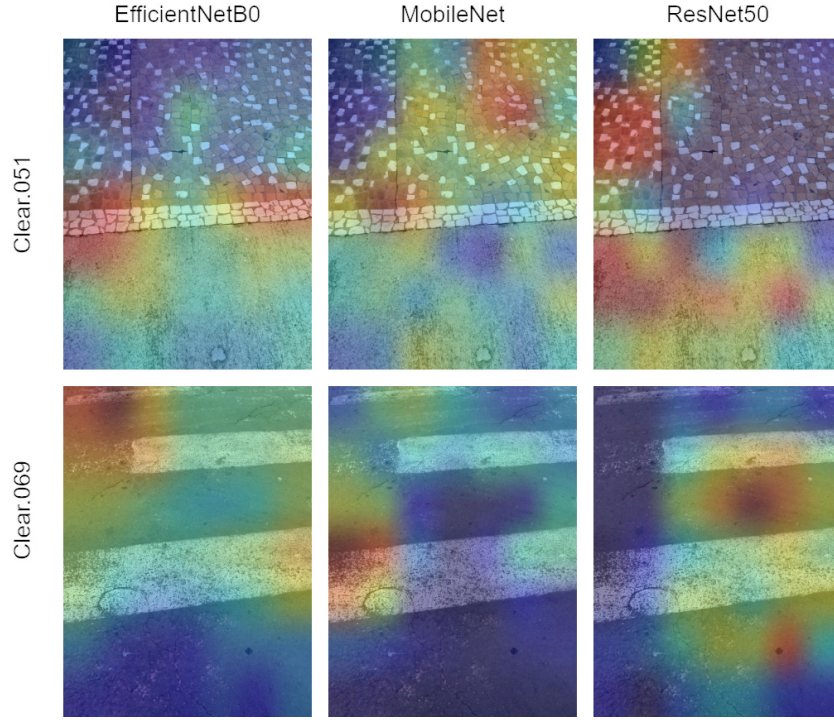


Figure 10: Heatmap of Misclassified Obstructed Paths (All Approaches).

the CNN EfficientNetB0, there was an emphasis on the change in the floor pattern, but the heatmap was more defined along the entire corridor path, which did not occur with the other approaches.

In Fig. 13, there are examples of images where at least three approaches made incorrect predictions; these images represent paths with obstacles.

In the image nonclear.000, the CNNs did not characterize the importance of the main obstacle, which is a staircase, but rather focused on a plant in the garden that does not necessarily represent a problem for the mobility of visually impaired individuals. In the image nonclear.161, there is a more precise focus by the CNN EfficientNetB0 on the obstacle, which is a small hole with the remainder of an iron pipe from a removed post, while the ResNet50 also maintains some focus on the obstacle and the MobileNet spreads its heat zone across the sidewalk.

Many images with incorrect classification are related to markings painted on the ground or to traffic signs: the approaches under study characterize these markings as obstacles. A total of 27 images were identified that char-

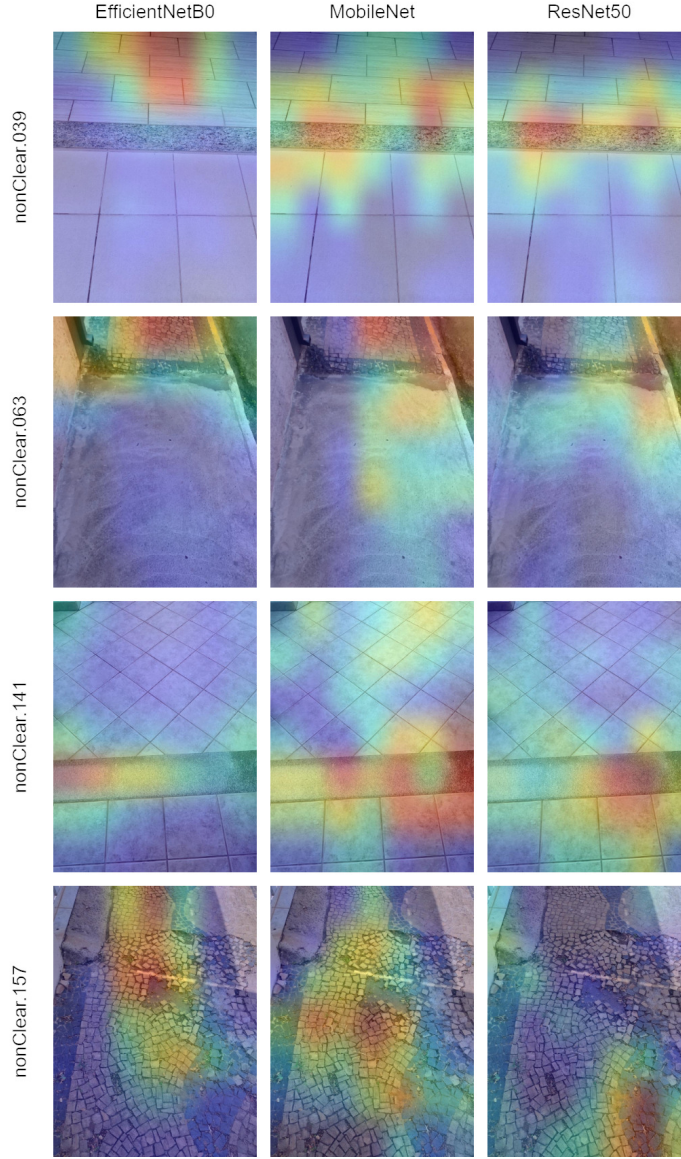


Figure 11: CNN Activation Heatmap for Obstructed Path Images with Errors in All Four Approaches.

acterize paths without obstacles with ground signage; among these images, the error rate was 59.29%, meaning 16 images were incorrectly classified by at least one of the studied approaches. This issue is illustrated in Fig. 14, where CNN activations focus on traffic markings.

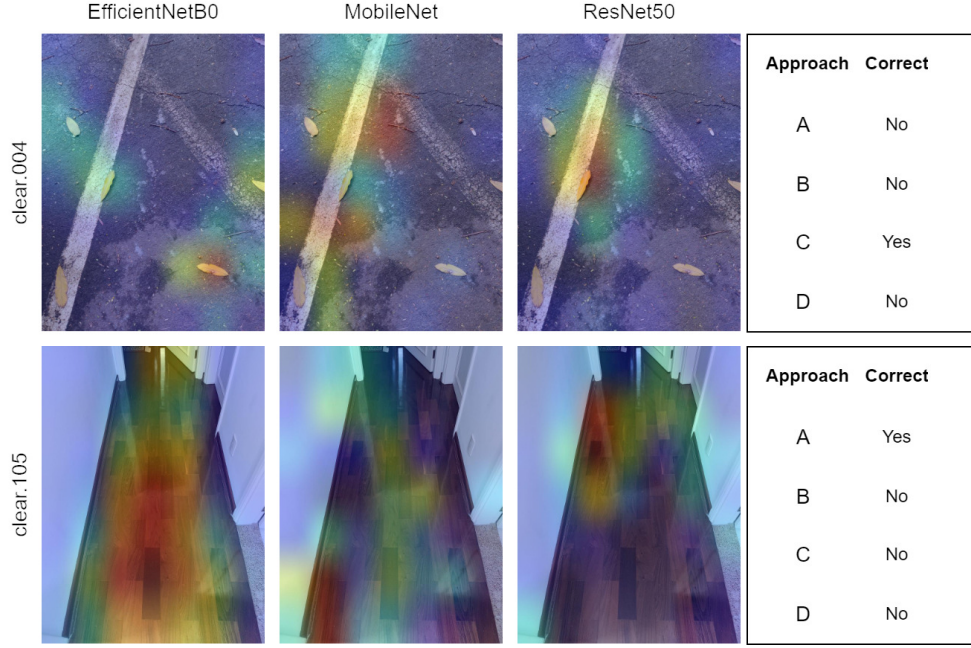


Figure 12: CNN Activation Heatmap for Path Without Obstacles Images with Errors in Three of the Four Approaches.

The focus of the CNN activations concentrates on ground signage, which does not represent an obstacle to visually impaired individuals. In Fig. 15, images with obstacles are observed; in both displayed images, the curb edge is presented as an obstacle to visually impaired individuals. In the image nonclear.033, although all approaches had their predictions correct, the focus of activation for CNNs EfficientNetB0 to a greater degree, and MobileNet, was on the ground signage, and only the CNN ResNet50 focused on the curb edge. Likely, the correct classification provided by the approaches was based on feature extraction that emphasizes traffic signage and not the real obstacle, which is the curb edge.

In the image nonclear.160 (Fig. 15), the CNN ResNet50 demonstrates a wider spread of attention points, but still focusing on traffic signage, and the CNNs EfficientNetB0 and MobileNet have activation points strongly related to traffic signage, again not noticing the curb edge.

The performance of the proposed approaches was assessed in relation to images featuring traffic signage, which can be viewed in Table 10. Approach A did not show much fluctuation in accuracy compared to the average ob-

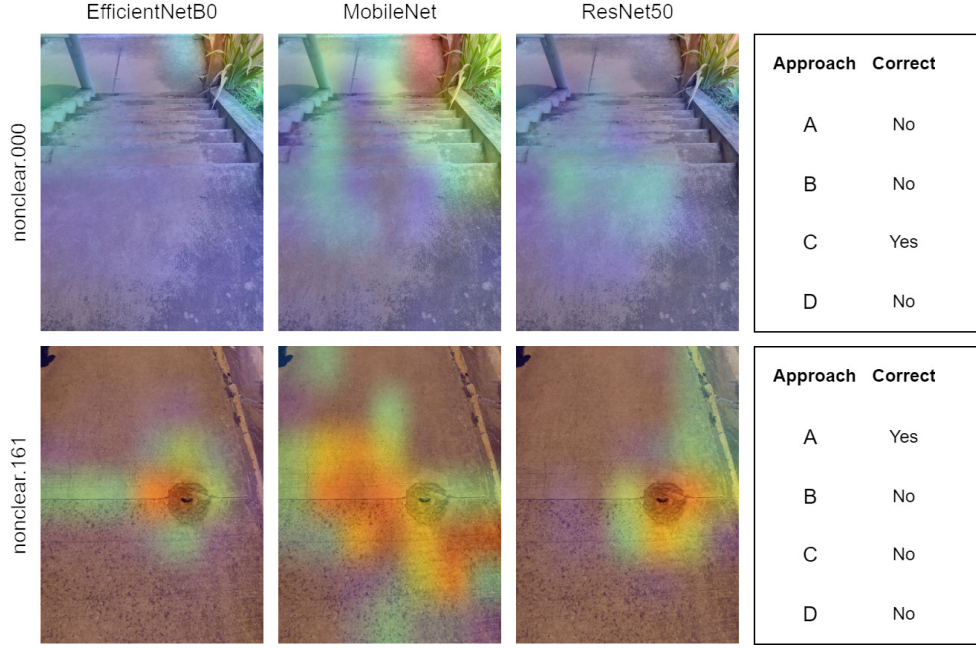


Figure 13: CNN Activation Heatmap for Obstructed Path Images with Errors in Three of the Four Approaches.

served in the other two tests conducted. In Approach C, an improvement in accuracy was observed compared to the individual test with all images and a decrease compared to the test with cross-validation (10-fold). In approaches B and D, a significant drop in accuracy was noted in the test with images featuring ground signage. This fact raises questions about whether the process of dimensionality reduction causes the loss of important information for the correct prediction of this type of image.

Table 10: Accuracy Comparison of Proposed Models Using LOOCV for Ground Signage Images, General LOOCV, and 10-fold Cross-Validation

Proposed Model	LOOCV Accuracy (Ground Signage Images)	LOOCV Accuracy	10-fold Accuracy
Approach A	0.9259	0.9474	0.9412
Approach B	0.4815	0.8713	0.9421
Approach C	0.8888	0.8421	0.9559
Approach D	0.7037	0.8509	0.9278

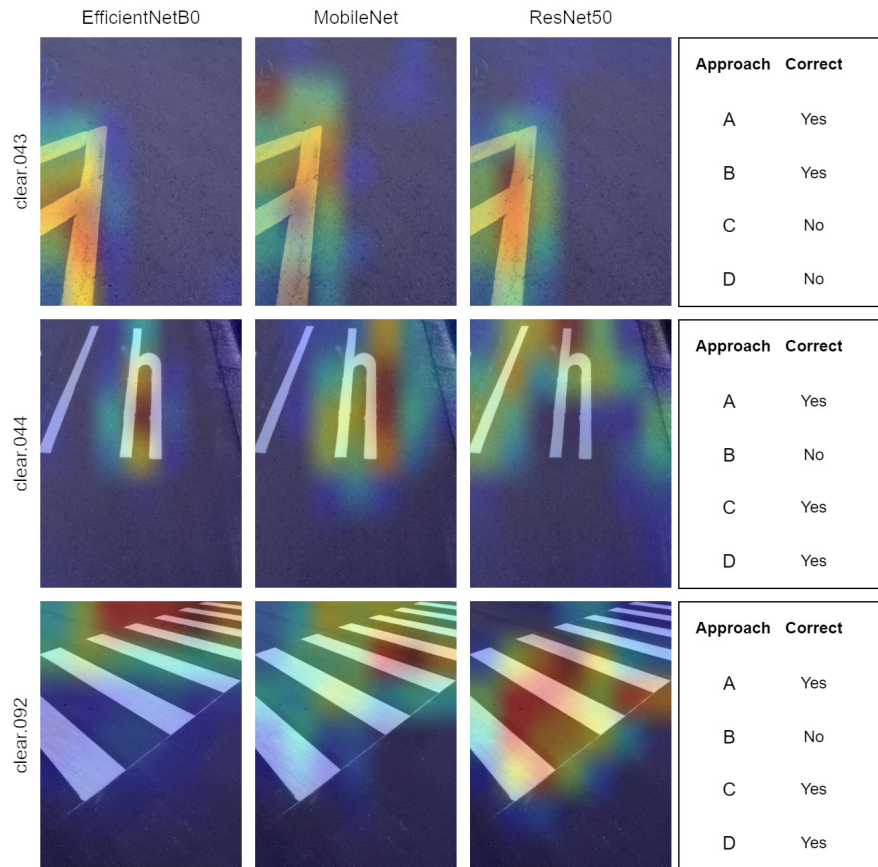


Figure 14: CNN Activation Heatmap for Clean Path Images with Ground Traffic Signage.

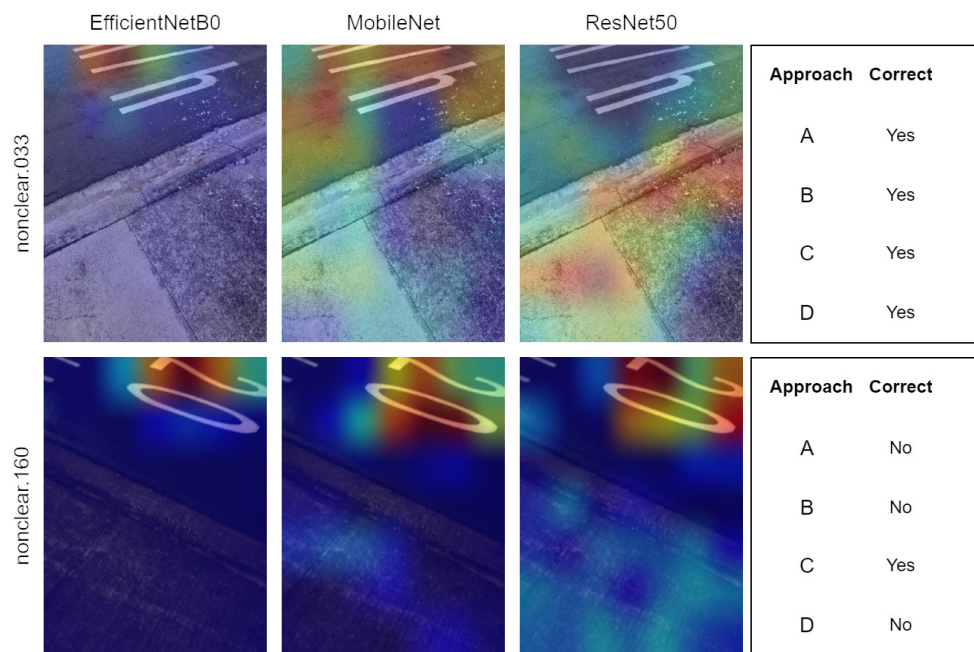


Figure 15: CNN Activation Heatmap for Obstructed Path Images with Ground Traffic Signage.

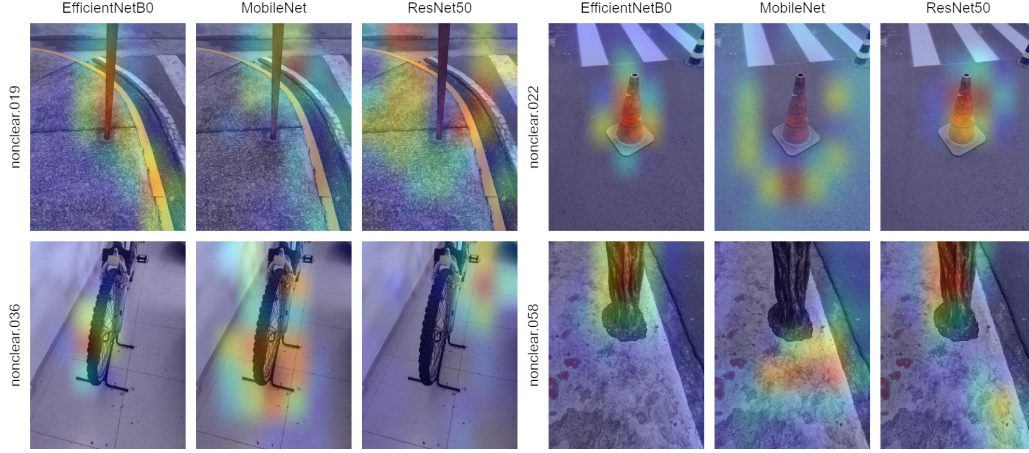


Figure 16: CNN Activation Heatmap for Image with No Classification Error.

Regarding the images without classification errors, a common characteristic is that these images depict everyday objects that are certainly included in the ImageNet database. Thus, due to the transfer learning applied to the CNNs used in this work, it likely facilitates the identification process, as evidenced by the heat zone always being on the objects in Fig. 16.

If we perform a comparison with the study published by Breve [27], where the aim was to optimize the performance of CNNs for classification within the image dataset, it can be observed that the best performance was achieved in a scenario of six instances of EfficientNetB4 combined with six instances of EfficientNetB0, with fine-tuning, and six instances of the CNN MobileNet, which were used as feature extractors. This approach from Breve’s study [27] resulted in an accuracy of 0.9655. In comparison with this study, the approach with the highest accuracy achieved 0.9559, and the most stable across all studied situations achieved a best result of an accuracy of 0.9474. It should be emphasized that in this study, the CNNs were used without any alterations to their structure, thus eliminating the need for training and solely utilizing transfer learning from the ImageNet database.

Additional Metrics and Computational Cost of the Recommended Model

Because accuracy alone is insufficient in a safety-sensitive task, the most stable configuration – Approach A (EfficientNetB0 + MobileNet with a linear SVM), recommended for deployment – was further characterized in terms of F1-score and ROC-AUC, computed per fold. Table 11 reports the median

values together with the first and third quartiles.

Table 11: Additional performance metrics for the recommended configuration (Approach A: EfficientNetB0 + MobileNet + linear SVM), over the 10 folds.

Metric	Q1 (25%)	Median	Q3 (75%)
Accuracy	0.9118	0.9412	0.9706
F1-score	0.9231	0.9412	0.9730
ROC-AUC	0.9118	0.9479	0.9688

The computational cost of this configuration was also measured per fold on the notebook used for the proposed approaches. The one-time feature extraction with the two CNNs took approximately 45.9 s for the entire 342-image dataset (about 134 ms per image), the training of the linear SVM required about 2.8 s, and classifier inference was approximately 6.7 ms per image. As the feature extraction relies on lightweight MobileNet-based architectures and the classification stage is computationally trivial, the pipeline is compatible with on-device execution, although smartphone-specific latency and memory footprint remain to be measured.

Limitations and Future Work

This work has limitations that delimit the scope of its conclusions. First, the evaluation relied on a single, modest-sized dataset (342 images from the VIA dataset); external validation on larger and independent datasets is required to assess generalisation. Second, the best technique of each approach was selected among thousands of combinations, which may optimistically bias the peak 10-fold accuracy (0.9559); the leave-one-out estimate of the most stable configuration (0.9474) should therefore be regarded as the more reliable performance indicator. Third, beyond accuracy, a fuller characterisation in terms of sensitivity and specificity is desirable, given the asymmetric cost of failing to detect an obstacle. Finally, the present study is an offline comparative analysis; the integration of the selected model into a smartphone application and its on-device evaluation – real-time inference, memory footprint, and a usability assessment with visually impaired users – constitute the next phase of this research.

7. Conclusion

This study explored the use of 25 CNN architectures as feature extractors, combined with dimensionality reduction and supervised classification,

for the detection of obstacles in environments navigated by visually impaired individuals. The models, applied without any retraining, showed competitive accuracy compared to fine-tuned alternatives. The best-performing model reached 0.9559 accuracy, and the most stable configuration across tests achieved 0.9474. A Nemenyi post-hoc analysis confirmed that the proposed approaches are statistically equivalent to the fine-tuned reference model, yet at a fraction of its computational cost: once the features have been extracted, classifier inference takes only about 6.7 ms per image. These results demonstrate that systems based exclusively on transfer learning can achieve high performance without significant computational costs. However, limitations were observed in interpreting floor signage, indicating potential for improvement through attention mechanisms or hybrid semantic segmentation. As an offline comparative study, this work establishes the feature-extraction pipeline and the selected model, providing the basis for the on-device deployment and evaluation phase outlined above.

Code and Data Availability

The dataset, the source code, and the complete experimental results that support the findings of this study are publicly available at <https://github.com/jeffersonarpasserini/Phd-Partial-Results>.

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